

POSTERIOR-TAIL SELECTION IN BAYESIAN ENSEMBLE WORLD MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

Bayesian and ensemble world models expose epistemic uncertainty, so they are often treated as a safer substrate for planning. This paper studies a failure in how that uncertainty is consumed: score each candidate trajectory with a posterior-sampled model member, execute the candidate with the largest sampled score, and the selector can convert epistemic uncertainty into a selected upper-tail residual. We call this failure *posterior-tail selection*. In a controlled latent dynamics benchmark with 5 posterior seeds and 225 candidate problems, sampled-posterior max selection at $N = 128$ reaches true return 5.567 and regret 10.888 to the in-set candidate oracle, while posterior-mean selection reaches true return 16.355 and regret 0.100. The selected sampled-posterior trajectory has posterior standard deviation 8.234, sample-over-mean residual 13.726, OOD mass 0.304, and hazard-tail selection rate 0.276. A calibration-aware lower-confidence selector reduces regret to 0.233 and tail selection to 0.004 while retaining most of the candidate-oracle value. The contribution is intentionally narrow: a mechanism, diagnostic benchmark, and repair for posterior-tail aggregation in Bayesian or ensemble world-model rollouts, not a claim of real-robot or large-scale benchmark dominance.

1 INTRODUCTION

World models let agents evaluate futures before acting. Modern systems learn latent dynamics for planning and policy learning (Hafner et al., 2019; 2020; 2023), use ensembles to represent model uncertainty (Chua et al., 2018; Lakshminarayanan et al., 2017), and often combine learned models with model-predictive control or offline policy optimization (Janner et al., 2019; Yu et al., 2020; Kidambi et al., 2020; Yu et al., 2021). A common deployment instinct is to spend more inference-time compute: sample N candidate action sequences, roll them out in the world model, and choose the best.

This paper asks what happens when the deployment score is not a posterior mean or lower confidence bound, but a separately sampled posterior value for each candidate. The rule resembles Thompson sampling locally, but the many-candidate maximum changes its behavior: a candidate can win because one plausible model member hallucinates high return in an epistemically uncertain region. The agent then executes a trajectory that is not robust under the posterior.

The phenomenon is related to reward-model overoptimization in language models (Gao et al., 2023; Jinnai et al., 2025; Huang et al., 2025; Yu et al., 2026; Hsu & Lei, 2026). The difference is architectural. In Bayesian world models, the error source is a learned dynamics posterior over imagined trajectories, and the diagnostic should measure trajectory OOD mass, ensemble dispersion, and posterior sample-over-mean gaps.

We contribute:

- the posterior-tail selection failure mode for Bayesian and ensemble world-model planning;
- a compact selected-residual proposition showing why a maximum over posterior-sampled scores exposes upper-tail model error;
- a deterministic latent dynamics benchmark for posterior-sampled and ensemble world-model selectors;

- selected-trajectory diagnostics for OOD mass, hazard mass, ensemble dispersion, sampled residual, and regret to the in-set candidate oracle;
- a calibration-aware lower-confidence repair that uses held-out residual calibration and test-time ensemble predictions only.

2 MECHANISM

Let $a_1, \dots, a_N$ be candidate action sequences. A Bayesian or ensemble world model gives each candidate a posterior distribution over imagined returns. Write a posterior sample score as

$$S_i = \mu_i + \epsilon_i, \quad (1)$$

where μ_i is the posterior mean and ϵ_i is the sampled model-member residual. The sampled-posterior max selector chooses $\hat{i} = \arg \max_i S_i$.

Diagnostic proposition. Suppose ϵ_i are independent centered residuals conditional on the candidate set, and suppose at least two candidates have nonzero residual variance. Then

$$\mathbb{E}[\epsilon_{\hat{i}}] \geq 0, \quad (2)$$

with strict inequality whenever residual variance can affect the argmax. Moreover, in equal-mean Gaussian residuals with variance σ^2 , the expected selected residual scales as $\sigma \mathbb{E}[\max_i Z_i]$, which grows like $\sigma \sqrt{2 \log N}$.

This is not a theorem about all learned controllers. It is a warning about the aggregation operator. Posterior sampling is useful for exploration when one sampled model is followed coherently. A maximum over independently sampled trajectory scores instead searches for posterior outliers.

Repair principle. Use a posterior-robust score:

$$L_i = \mu_i - \beta \sigma_i, \quad (3)$$

where σ_i is ensemble standard deviation and β is fitted from held-out calibration candidates as a high quantile of $|\mu_i - V_i|/(\sigma_i + \epsilon)$. This is deliberately simple: the paper tests whether the selected-tail failure is visible enough that a scalar lower-confidence rule removes it.

3 BENCHMARK

We build a two-dimensional latent control system with position and velocity. The true dynamics reward approaching a goal region and penalize entering a high-position hazard. Candidate action sequences are smooth random controls with occasional high-excursion probes. The posterior ensemble is accurate near the training-like region but uncertain about hazard strength and terminal value in high-position regions.

For every problem instance, we draw a shared candidate set of size 128 and evaluate smaller budgets by prefixing the same set. This makes curves across N comparable. The selectors are:

- **Posterior mean:** choose the candidate with largest ensemble mean.
- **Sampled-posterior max:** choose the largest score after sampling one ensemble member per candidate.
- **Thompson-member max:** choose a single model member and rank all candidates by that member.
- **Mean + std:** an optimistic upper-confidence selector.
- **Quantile and calibrated LCB:** posterior-pessimistic repairs.
- **Oracle:** choose the highest true return within the same candidate set.

The full sweep uses 5 posterior seeds and 225 candidate problems. The code writes all detailed rows to `results/full/results.csv`; the paper figures are generated from `results/full/summary.csv`.

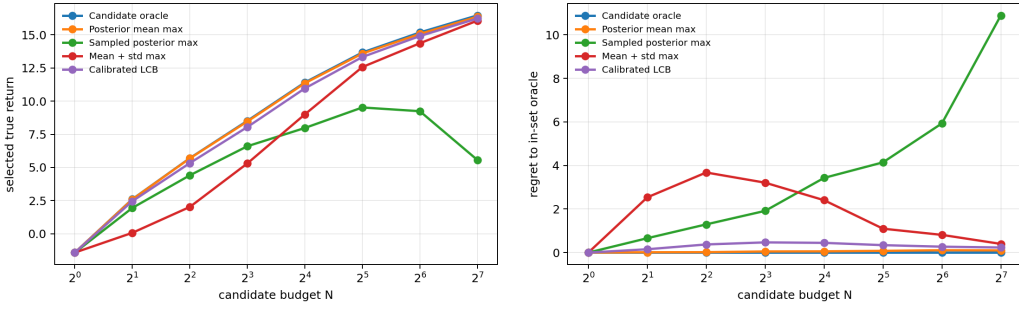


Figure 1: Left: selected true return. Right: regret to the best true-return candidate in the same sampled set. Sampled-posterior max selection degrades at large N despite the candidate set containing strong trajectories.

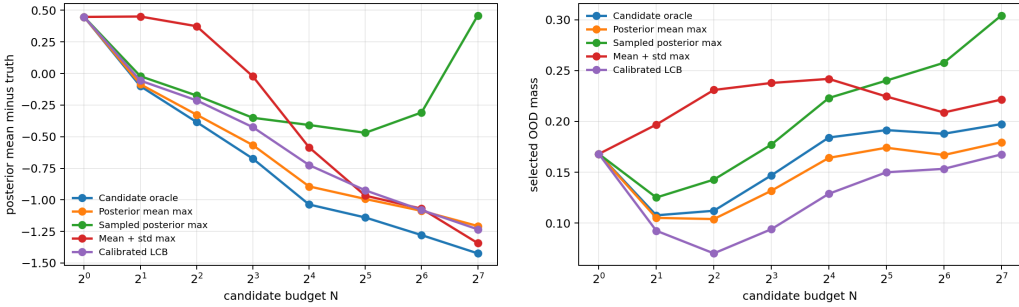


Figure 2: Selected-trajectory diagnostics. Sampled-posterior max selection chooses candidates with higher OOD mass and large sampled-score residuals over the posterior mean.

4 RESULTS

Figure 1 shows the main result. At $N = 128$, sampled-posterior max reaches true return 5.567 with regret 10.888. Posterior mean reaches 16.355 with regret 0.100, and the candidate oracle reaches 16.455. The failure is therefore not lack of good candidates; it is the posterior-sampled selection rule.

Figure 2 explains the mechanism. The posterior-sampled winner has selected OOD mass 0.304, posterior standard deviation 8.234, and sampled score 13.726 above its posterior mean at $N = 128$. In contrast, calibrated pessimism has selected OOD mass 0.168 and tail-selection rate 0.004. The lower-confidence repair obtains true return 16.223 and regret 0.233.

Posterior mean is slightly better than calibrated pessimism in this benchmark. This is a useful boundary: the paper is not claiming that pessimism always improves over posterior means. It claims that if a deployment system uses posterior-sampled or upper-tail ensemble scores for many-candidate planning, it should audit selected residuals and consider lower-confidence aggregation.

5 RELATED WORK

World models and ensemble planning. PlaNet and Dreamer learn latent dynamics for imagination and control (Hafner et al., 2019; 2020; 2023). PETS introduced probabilistic ensembles with trajectory sampling for sample-efficient continuous control (Chua et al., 2018). MBPO studies when short model rollouts are useful for policy optimization (Janner et al., 2019). These works motivate the architecture, while our focus is the inference-time aggregation operator over posterior-sampled candidate scores.

Pessimism and model exploitation. MOPO, MOREL, COMBO, RAMBO, and CQL use pessimism or conservatism to avoid model or value overestimation in offline RL (Yu et al., 2020; Kidambi

et al., 2020; Yu et al., 2021; Rigter et al., 2022; Kumar et al., 2020). Our repair is much simpler and narrower: a calibrated lower-confidence score for a candidate selector.

Bayesian uncertainty. Deep ensembles, MC dropout, bootstrapped value functions, randomized priors, and posterior-sampling RL all provide uncertainty tools (Lakshminarayanan et al., 2017; Gal & Ghahramani, 2016; Osband et al., 2016; 2018; Fan et al., 2021). Recent robotic world-model work also emphasizes epistemic uncertainty for deployment (Li et al., 2025). The current diagnostic asks how that uncertainty is consumed by many-candidate maximum selection.

Inference-time overoptimization. Reward-model overoptimization and inference-time alignment work show that many-candidate selection can reward-hack imperfect learned scorers (Gao et al., 2023; Jinnai et al., 2025; Huang et al., 2025; Yu et al., 2026; Hsu & Lei, 2026). We import the operator-level warning but specialize it to Bayesian ensemble world models.

6 LIMITATIONS

This is a first-pass controlled benchmark. It does not show real-robot results, MuJoCo results, or large video world-model experiments. The synthetic posterior is hand-designed to expose an identifiable failure mode. The proposition assumes centered residuals and independence that are only approximations in learned dynamics. The repair is a scalar LCB rather than a conformal or task-adaptive uncertainty method. These limits are deliberate: the paper should be read as a falsifiable diagnostic and artifact, not as a final deployment recipe.

7 REPRODUCIBILITY AND ETHICS

The repository includes code, tests, deterministic results, figures, literature notes, hostile-prior analysis, and a final audit. The benchmark is synthetic and has no human-subject data. The main safety implication is cautionary: increasing inference-time compute can amplify model error when the learned scorer is optimized too aggressively.

REFERENCES

- Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems*, 2018.
- Xidong Fan, Yifei Ming, Yuchen Zhang, and Huazhe Zhang. Model-based reinforcement learning for continuous control with posterior sampling. In *International Conference on Machine Learning*, 2021.
- Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning*, 2016.
- Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, 2023.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning*, 2019.
- Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations*, 2020.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.
- Daniel Hsu and Qi Lei. Bridging optimism and pessimism in inference-time alignment. *arXiv preprint arXiv:2603.06797*, 2026.

- Audrey Huang, Adam Block, Qinghua Liu, Nan Jiang, Dylan J. Foster, and Akshay Krishnamurthy. Is best-of-n the best of them? coverage, scaling, and optimality in inference-time alignment. In *Conference on Learning Theory*, 2025.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems*, 2019.
- Yuu Jinnai, Tetsuro Morimura, Kaito Ariu, and Kenshi Abe. Regularized best-of-n sampling with minimum bayes risk objective for language model alignment. In *Annual Conference of the North American Chapter of the Association for Computational Linguistics*, 2025.
- Rahul Kidambi, Aravind Rajeswaran, Praneeth Netrapalli, and Thorsten Joachims. Morel: Model-based offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020.
- Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine. Conservative q-learning for offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, 2017.
- Chenhao Li et al. Learning robotic policies without a physics simulator. *arXiv preprint arXiv:2504.16680*, 2025.
- Ian Osband, Charles Blundell, Alexander Pritzel, and Benjamin Van Roy. Deep exploration via bootstrapped dqn. In *Advances in Neural Information Processing Systems*, 2016.
- Ian Osband, John Aslanides, and Albin Cassirer. Randomized prior functions for deep reinforcement learning. In *Advances in Neural Information Processing Systems*, 2018.
- Marc Rigter, Bruno Lacerda, and Nick Hawes. Rambo-rl: Robust adversarial model-based offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2022.
- Tianhe Yu, Garrett Thomas, Lantao Yu, Stefano Ermon, James Zou, Sergey Levine, Chelsea Finn, and Tengyu Ma. Mopo: Model-based offline policy optimization. In *Advances in Neural Information Processing Systems*, 2020.
- Tianhe Yu, Aviral Kumar, Rafael Rafailov, Aravind Rajeswaran, Sergey Levine, and Chelsea Finn. Combo: Conservative offline model-based policy optimization. In *Advances in Neural Information Processing Systems*, 2021.
- Zhuohao Yu, Zhiwei Steven Wu, and Adam Block. From curiosity to caution: Mitigating reward hacking for best-of-n with pessimism. *arXiv preprint arXiv:2604.04648*, 2026.

A CHECKLIST

- **Claims:** The main claims are limited to the synthetic benchmark and the stated diagnostic proposition.
- **Code:** The source package is under `src`; experiments are in `experiments/run_benchmark.py`.
- **Data:** All data are generated synthetically from fixed seeds.
- **Compute:** The full sweep runs on CPU in roughly one minute on the development machine.
- **Prior work:** The repo includes a 100+ entry related-work matrix and a hostile-prior document.

B ADDITIONAL DIAGNOSTICS

At $N = 128$, sampled-posterior max selects hazard-tail trajectories with rate 0.276, while calibrated pessimism selects them with rate 0.004. The method is not an oracle: it uses held-out calibration candidates to fit β and never reads test-candidate true returns.